Introduction

What does it take for a robot to perform a task that a human can do almost without thinking?

Pick up a part. Identify it. Move it. Put it in the right place. Sounds simple. For a humanoid robot, it is anything but.

My name is Whilliam and my name is Oskar

And we have been chosen to present our project, which investigates whether a humanoid robot can perform an industrial kitting task autonomously. We are using the Unitree G1 education mode and focusing on the complete pipeline from detecting objects to picking and placing them in the correct locations.

Humanoid robots have become increasingly capable of walking and performing large sweeping motions. However, industrial tasks often require much more precision than what is currently available. To achieve what a human could, the robot needs to identify an object, determine where it is, plan how to reach it, grasp it and place it accurately. If it can perform these tasks reliably, it can be used in industrial settings without requiring costly hardware changes to the machine.

The previous master's thesis showed us that this concept is possible. They used telescoping to train a VLA or a “vision language action model” that controlled the whole pipeline. However, the system was slow and unreliable because it was trained on human movement patterns that led to jitter and heavy computation. So instead of starting completely from scratch, we want to build on that work. Our idea is to use AI where it is strongest, which is understanding and detecting objects and use more deterministic methods for the actual arm movement, where predictability and precision are more important.

To reach the goal and requirements of this project the robot needs to be able to detect objects and classify them. Then, it has to estimate the position as well as orientation of the object so it can reach the object and pick it up.  Finally, identifying the correct location on the kitting tray and placing the object correctly by manipulating the objects 6 DOF.

Lets talk about the robot in question, the unitree G1 education. It is equipped with a stereo-vision camera in combination with a 3D lidar for vision sensors. The robot is bundled with a set of inspire ftp dexterous hands with about six pressure sensors integrated within. This would allow the robot to sense gripping strength and “feel” if the object is in the hand or dropped. This is important since the assignment needs precision work. The robot is controlled by a  NVIDIA Jetson Orin 100 TOPS built for high speed, interface support for the multiple sensors. With this, the hardware limitations that the project would entail is eliminated and the focus will be on the software.

The initial ideas for robotic control are YOLO, FoundationPose, MoveIt2 and Pinocchio

YOLO (You Only Look Once) is a real-time object detection model designed to quickly identify and locate objects within an image. By processing the entire image in a single pass, we can detect multiple objects simultaneously while providing their class, location, and confidence score. This combination of speed and accuracy makes it well suited for robotics, where objects need to be detected and located to support tasks such as grasping and manipulation.

FoundationPose is a way to figure out the object's 6DOF in space, thus being able to figure out the correct plan for manipulation of the object. It does this by comparing the detected object to a 3d model. This would give us both the xyz of the object as well as the rotational position.

For movement calculations, we plan on combining MoveIt 2 and Pinocchio. MoveIt2 is a framework that helps the robot figure out how to move its arm from one position to another. It takes information about the robot and its surroundings, then it would calculate a suitable path for the robot to follow. This in combination with…
Pinocchio which would give the robot a mathematical understanding of its own body. With this framework the robot is represented as a chain of joints and links, with each joint describing how one part can move relative to another.
Pinocchio and MoveIt 2 can work together to handle the robot's kinematics and motion planning which would be assisted by the vision sensor.

Finally, IsaacLab is an open source simulation framework that is made for testing and simulating robot behaviour in a virtual environment. This system includes realistic physics models, collision handling and sensor simulation which makes it suitable for our project.

Moving on to our project plan, our project can be divided into three separate parts. Firstly, the perception part where the objects are detected and the object position is estimated. This part also includes identifying the right place on the kitting tray. Secondly, there is the manipulation part which includes reaching the object, grasping and placing it. Thirdly, we have the integration part containing how we implement our work, such as the architecture of ROS and the communication between all parts.

To reiterate, the expected outcome would include the previous work modules working in tandem. Going from detection to movement to pick up the object in a way to move it without dropping the object and placing it down in the correct tray. If it does not succeed, the robot should be able to recognise this and continue to attempt the kitting task until it determines it has placed it correctly.

Thus, the demoplan for 3rd of december is to demonstrate the robot completing a kitting task in an ideal and easy scenario. This is so that we can demonstrate meaningful progress even if every advanced feature has not been completed. The primary target is the complete autonomous task, but individual modules i.e detection, movement, manipulation, will also provide evidence of our development.

With the previous master thesis work as a basis, and our plan to divide up the work between modules, there is a high likelihood that by discarding the idea of an all encompassing VLA, we will be able to complete the goal in time. So despite the myriad of design obstacles ahead, by combining these modules and with continued research, it is within my belief that the future of robotics will allow for individual robotic agents to complete tasks that a human can do almost without thinking. FIN